

MATH I

Week 3 (Fall 2026)

Practical Session Exercises

Exercises 2-8 were developed by G. Neofotistos in conversations with OpenAI model 5.6 Sol (Sept 2026)

Engineering Problems on Limits

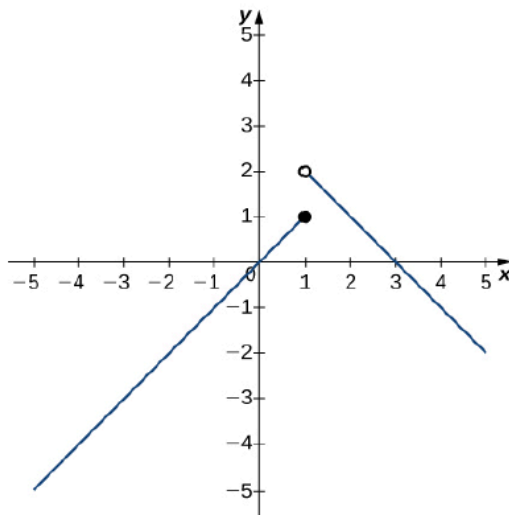
Students should:

- show the mathematical procedure;
- include appropriate physical units (do not forget the units!);
- interpret the result in engineering terms;
- state when a limit does not exist.

Exercise 1:

This exercise comprise problems 50-54 from Chapter 1 of textbook.

In the following exercises, use the following graph of the function $y = f(x)$ to find the values, if possible. Estimate when necessary.



51. $\lim_{x \rightarrow 1^+} f(x)$

54. $f(1)$

52. $\lim_{x \rightarrow 1} f(x)$

50. $\lim_{x \rightarrow 1^-} f(x)$

53. $\lim_{x \rightarrow 2} f(x)$

Exercise 2: One-sided limits

Consider

$$f(x) = \begin{cases} 2x + 1, & x < 2, \\ 7 - x, & x \geq 2. \end{cases}$$

Find:

$$\lim_{x \rightarrow 2^-} f(x), \quad \lim_{x \rightarrow 2^+} f(x), \quad \lim_{x \rightarrow 2} f(x), \quad f(2).$$

Explain why the two-sided limit does or does not exist.

Exercise 3: Calibration near a missing measurement

The output of a temperature sensor is modelled by

$$V(T) = \frac{T^2 - 2500}{T - 50}, \quad T \neq 50,$$

where T is measured in degrees Celsius and V in millivolts.

1. Explain why $V(50)$ is undefined.
2. Estimate

$$\lim_{T \rightarrow 50} V(T).$$

3. Explain how the limit could be used to replace the missing calibration value.
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Exercise 4: A switched control system

The voltage supplied by a controller is

$$V(t) = \begin{cases} 2t + 1, & t < 4, \\ 12 - t, & t \geq 4. \end{cases}$$

Determine:

$$\lim_{t \rightarrow 4^-} V(t), \quad \lim_{t \rightarrow 4^+} V(t), \quad \lim_{t \rightarrow 4} V(t), \quad V(4).$$

Is there an abrupt voltage change at $t = 4$? Explain what this could mean for the controlled device (*think engineering-ly!!*)

Exercise 5: Stress–strain behaviour

Introduction: **Stress** describes the internal force acting per unit area within a material when an external load is applied, commonly written as $\sigma = F/A$ and measured in pascals (Pa). **Strain** describes the resulting relative deformation: for axial loading, $\epsilon = \Delta L/L_0$, and it is dimensionless. The general stress–strain relationship depends on the material and loading conditions and may be written as $\sigma = f(\epsilon)$. In the linear-elastic region, many materials approximately obey **Hooke's law**, $\sigma = E\epsilon$, where E is Young's modulus; after the elastic limit, the relationship may become nonlinear and involve permanent (plastic) deformation, eventually leading to fracture.

Statement of the problem:

An idealized material has the stress–strain relationship

$$\sigma(\epsilon) = \begin{cases} 200\,000\epsilon, & \epsilon < 0.002, \\ 400 + 10\,000(\epsilon - 0.002), & \epsilon \geq 0.002, \end{cases}$$

where stress σ is measured in MPa and strain ϵ is dimensionless.

1. Find

$$\lim_{\epsilon \rightarrow 0.002^-} \sigma(\epsilon).$$

2. Find

$$\lim_{\epsilon \rightarrow 0.002^+} \sigma(\epsilon).$$

3. Does the two-sided limit exist?
 4. Does the model predict a sudden jump in stress at the transition?
 5. Sketch the stress–strain graph near $\varepsilon = 0.002$.
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Exercise 6: Long-term RC-circuit response

Introduction: A **capacitor** is an electrical component that stores electric charge and energy in the electric field between two conducting plates separated by an insulating material. Its capacitance C relates the stored charge Q to the voltage V by $Q = CV$. When a capacitor is connected to a DC source V_s through a resistor R , it charges gradually rather than instantaneously: the voltage across the capacitor is $V_C(t) = V_s (1 - e^{-t/(RC)})$, assuming it is initially uncharged. Thus, $V_C(0) = 0$, and the voltage approaches V_s asymptotically as $t \rightarrow \infty$; the product $\tau = RC$, called the *time constant*, determines how quickly the capacitor charges.

Statement of the problem:

The voltage across a charging capacitor is

$$V(t) = V_s (1 - e^{-t/(RC)}),$$

where V_s is the supply voltage.

Let

$$V_s = 12 \text{ V}, \quad R = 10\,000 \, \Omega, \quad C = 100 \, \mu\text{F}.$$

1. Evaluate

$$\lim_{t \rightarrow 0^+} V(t).$$

2. Evaluate

$$\lim_{t \rightarrow \infty} V(t).$$

3. Interpret both limits physically.
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Exercise 7: Terminal velocity

The velocity of a falling object is modelled by

$$v(t) = v_T (1 - e^{-kt}),$$

where $v_T > 0$ is the terminal velocity and $k > 0$.

1. Find $v(0)$.
2. Evaluate

$$\lim_{t \rightarrow \infty} v(t).$$

3. Explain why this model is more realistic than a model in which velocity increases indefinitely (*think with common sense!*).
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Exercise 8: Data-transfer rate

The effective data-transfer rate of a communication channel is modelled by

$$R(n) = \frac{100n}{n + 20}$$

megabits per second, where n represents the allocated resource level.

1. Calculate $R(10)$, $R(100)$ and $R(1000)$.
2. Evaluate

$$\lim_{n \rightarrow \infty} R(n).$$

3. Explain why increasing n eventually produces diminishing improvements.
 4. Identify the maximum theoretical transfer rate predicted by the model.
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